ICTD COURSE CASE STUDY — APPENDIX

Answering the Five RCT Research Questions

Deck 2 of 2 — hypothesis tests, index construction, and mediation analysis for the Commons Connect randomized evaluation

Companion to Deck 1 (Commons Connect × the six ICTD topics). This deck expands Topic 5 — Impact Evaluation.

July 2026

Roadmap: From Question to Evidence

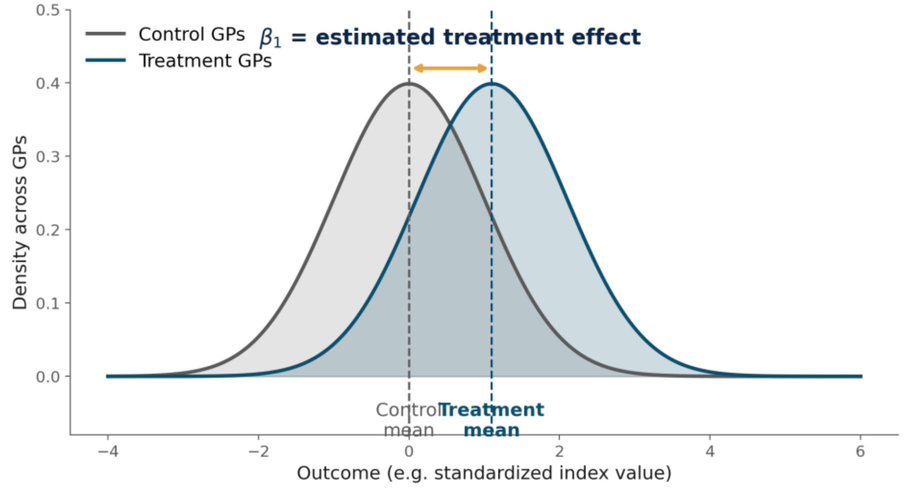
Methodology	How any RQ becomes a hypothesis test; controls; indices; multiple testing
RQ1 – RQ4	One index-construction slide + one hypothesis-test slide per research question
RQ5	Mechanism: does the effect run through information, transparency, or bargaining power?
Wrap-up	The full pipeline in one diagram, and discussion questions

From Research Question to Hypothesis Test

- Every RQ ultimately reduces to one test: does an outcome differ, on average, between treatment and control GPs, once assigned by randomization?
- This is estimated as a regression rather than a plain two-sample test, so that pre-treatment controls and clustered standard errors come along automatically:

$$Y_{ig} = \beta_0 + \beta_1 T_g + \boldsymbol{\beta}_2' X_{ig} + \varepsilon_{ig}$$

- $T_g = 1$ if GP g was randomly assigned to receive Commons Connect this cycle, 0 otherwise. β_1 is the intent-to-treat (ITT) effect — the object of interest for every RQ.



Baseline Controls: What Goes Into X_ig

Household / individual level (i)

Variable	Example question / data source
caste_ST	Household roster: "What is the caste/tribe category of the household head?" (SC/ST/OBC/General)
land_ha	Household roster: "How many hectares of land does your household own or operate?"
wealth_index	Constructed from a durable-asset ownership checklist (first principal component)
dist_gp_km	GIS-computed: straight-line distance from hamlet centroid to GP office
base_attendance	"Did you personally attend last year's Gram Sabha planning meeting?" (Y/N)

Gram Panchayat level (g)

Variable	Example question / data source
base_NRM_assets	Admin: pre-treatment count of existing water/NRM assets (Commons Connect baseline geotagging)
pop_SCST_share	Census / GP records: total population and SC/ST population share
agroclim_zone	GIS classification: agro-climatic / rainfall zone
CFR_recognized	GP records: "Does this GP have recognized Community Forest Rights?" (Y/N)
mgnrega_util	MGNREGA MIS: prior-year spending ÷ allocated budget

Outcome Variables: What goes into Y_ig

Variable (Y)	Level	Question asked / how data is picked up	Role
pct_SCST_demands	**g — GP**	**Admin: (# approved works geotagged to SC/ST-majority hamlets) ÷ (total approved works) per GP**	**Primary**
pct_women_beneficiary	g — GP	Admin: "Who is the primary beneficiary of this work?" (M/F/Joint) — share coded F or Joint	Secondary
SCST_attendance_share	g — GP	Enumerator headcount: "Of attendees present, how many self-identify as SC/ST?" ÷ total attendees	Secondary
felt_heard_SCST	i — Household	"My hamlet's needs were seriously considered when the GP decided what to build." (asked of SC/ST respondents)	Secondary

g = measured once per Gram Panchayat (repeats across every household in that GP).

i = measured separately for each household/respondent.

Sample Dataframe

Illustrative — not real data. *One row per household (i), flattened within each GP (g). Teal headers = X_ig controls, gold headers = Y_ig outcomes.*

gp_id	hh_id	T_g	pop_SCST_share% (g)	caste_ST (i)	pct_SCST_demands (g)	felt_heard_SCST (i)
GP014	HH01	1	0.52	ST	0.42	4
GP014	HH02	1	0.52	SC	0.42	3
GP014	HH03	1	0.52	OBC	0.42	4
GP007	HH01	0	0.49	ST	0.24	3
GP007	HH02	0	0.49	SC	0.24	2
GP007	HH03	0	0.49	General	0.24	3

Notice: every "(g)" column repeats identically across all 3 households within GP014, and again within GP007 — because they are GP-level facts, not household facts. Only the "(i)" columns vary row to row.

Building a Summary Index from Multiple Outcomes

- Each RQ has several plausible outcome variables. Rather than eyeballing four separate p-values, standardize each against the control group, then average into one domain index (Anderson 2008; Kling, Liebman & Katz 2007).

Step 1 — standardize each raw outcome to a z-score:

$$Z_{ig}^{(k)} = \frac{Y_{ig}^{(k)} - \overline{Y}_{\text{control}}^{(k)}}{SD_{\text{control}}^{(k)}}$$

Step 2 — average the K standardized outcomes into one index:

$$\text{Index}_{ig} = \frac{1}{K} \sum_{k=1}^{K} Z_{ig}^{(k)}$$

Interpretation: a 0.2 SD increase in the index means treatment-GP respondents scored, on average, 0.2 standard deviations higher across this whole family of outcomes — not just on any single item.

Outcome 1
(raw)

Outcome 2
(raw)

Outcome 3
(raw)

Standardized (z-scores)

Domain Index

The Hypothesis Test

Mapping the dataframe onto the regression:

Regression term	Dataframe column(s)
Y_ig (outcome)	representation_ig = z-avg(pct_SCST_demands, pct_women_beneficiary, SCST_attendance_share, felt_heard_SCST)
T_g (treatment)	T_g
X_ig (controls)	pop_SCST_share%, caste_ST + rest of household/GP controls (Methodology, Slide 4)

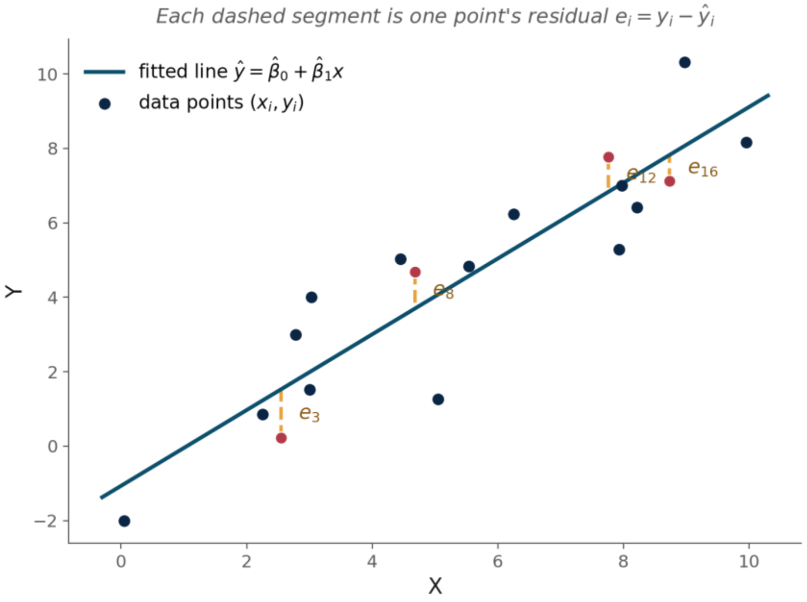
$$\mathrm{Representation}_{ig} = \beta_0 + \beta_1\, T_g + \boldsymbol{\beta}_2' X_{ig} + \varepsilon_{ig}$$

Feeding the dataframe in gives (illustrative, not real data):

Outcome	β_1 (est.)	SE	p
Representation_index (primary)	**0.19**	**0.08**	**0.017**

How Does Linear Regression Work?

- Every β_1 comes from fitting a straight line through data by least squares. Zooming into the mechanics with a generic X and Y (the logic is identical whether X is T_g or a covariate): $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \qquad e_i = y_i - \hat{y}_i$
- OLS chooses β_0, β_1 to minimize the average squared residual — the Mean Squared Error: $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n}\sum_{i=1}^{n} e_i^2$



This is exactly what the figure shows: the gold dashed segments are four e_i's, and the fitted line is the one position where the average of all e_i² (across every point, not just these four) is as small as possible.

R^2 — how much variance is explained

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} = 1 - \frac{SS_{res}}{SS_{tot}}$$

Fraction of Y's total variation the line accounts for. $R^2=1$: points sit exactly on the line. $R^2=0$: the line does no better than guessing $\bar{y}$ every time.

SE of the regression — typical residual size

$$SE_{\text{regression}} = \sqrt{\frac{1}{n-k}\sum_{i=1}^{n} e_i^2}$$

Roughly how far a typical point falls from the line, in Y's units (n−k adjusts for the k parameters used to fit it). The building block for β_1's own SE, generalized to clustering next.

How Are SE and p-values Computed?

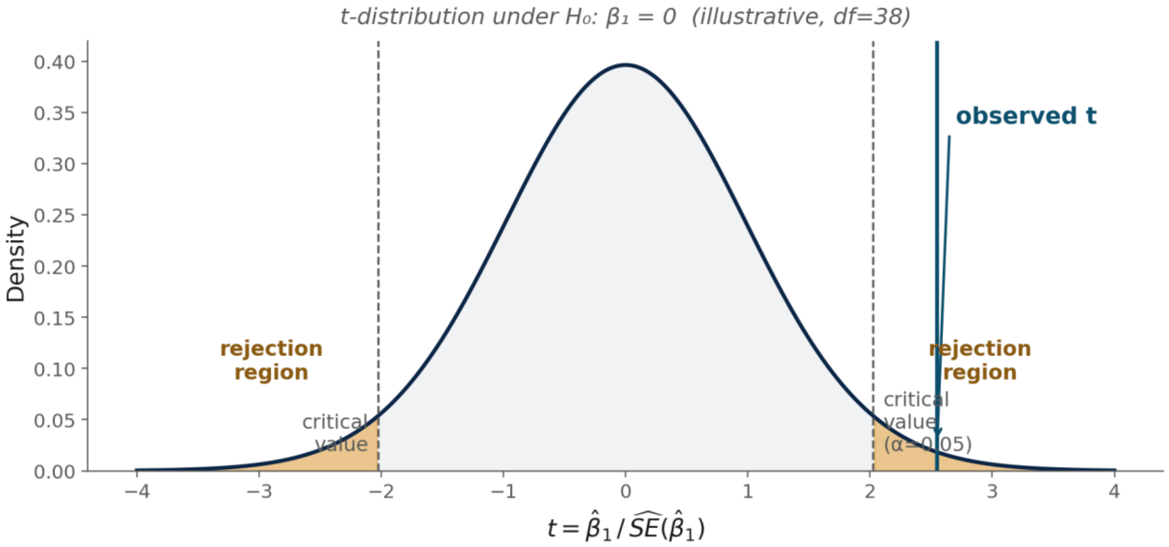
- The residual SE from the previous slide generalizes into a standard error for β_1 itself. With clustered data (households nested in GPs), that generalization is the cluster-robust "sandwich" formula:

- This SE turns β_1 into a test statistic, then a p-value:

$$t = \frac{\hat{\beta}_1}{\widehat{SE}(\hat{\beta}_1)}$$

$$p = 2 \times \Pr(T_{df} > |t|)$$

$p < 0.05$ means: if the true effect were zero, an estimate this far from zero would occur less than 5% of the time by chance alone. This is exactly what every β_1, γ, and α_1 in this deck is tested against — this slide is not repeated for RQ2–RQ5.



Outcome Variables: What goes into Y_ig

Variable (Y)	Level	Question asked / how data is picked up	Role
pct_NRM_climate_assets	**g — GP**	**Admin: work-type field tagged against a fixed NRM/climate-resilient category list; share of approved works so tagged**	**Primary**
site_suitability_score	g — GP	Tool output: CLART/recharge-potential score (0–100) queried at the exact lat-long of the sanctioned site, averaged per GP	Secondary
asset_functional_1yr	g — GP	"Is this structure currently holding/diverting water as designed?" (Yes/Partially/No), 1-year revisit, share per GP	Secondary

g = measured once per Gram Panchayat (repeats across every household in that GP).
i = measured separately for each household/respondent.

Sample Dataframe

Illustrative — not real data. *One row per household (i), flattened within each GP (g). Teal headers = X_ig controls, gold headers = Y_ig outcomes.*

gp_id	hh_id	T_g	base_NRM_assets (g)	land_ha (i)	pct_NRM_climate_assets (g)
GP014	HH01	1	5	1.2	0.55
GP014	HH02	1	5	0.8	0.55
GP014	HH03	1	5	2.1	0.55
GP007	HH01	0	6	1.5	0.30
GP007	HH02	0	6	0.9	0.30
GP007	HH03	0	6	1.8	0.30

Notice: every "(g)" column repeats identically across all 3 households within GP014, and again within GP007 — because they are GP-level facts, not household facts. Only the "(i)" columns vary row to row.

The Hypothesis Test

Mapping the dataframe onto the regression:

Regression term	Dataframe column(s)
Y_ig (outcome)	assetquality = z-avg(pct_NRM_climate_assets, site_suitability_score, asset_functional_1yr)
T_g (treatment)	T_g
X_ig (controls)	base_NRM_assets, land_ha + rest of household/GP controls (Methodology, Slide 4)

$$\mathrm{AssetQuality}_{ig} = \beta_0 + \beta_1 T_g + \boldsymbol{\beta}_2' X_{ig} + \varepsilon_{ig}$$

Feeding the dataframe in gives (illustrative, not real data):

Outcome	β_1 (est.)	SE	p
AssetQuality_index (primary)	**0.27**	**0.09**	**0.003**

Outcome Variables and Index Construction

Variable (Y)	Level	Question asked / how data is picked up	Role
attendance_share	**g — GP**	**Sign-in sheet headcount ÷ GP's registered adult population**	**Primary**
speaking_diversity_index	g — GP	Observer log: "How many distinct hamlets/groups had at least one member speak?" ÷ total hamlets	Secondary
deliberation_quality	g — GP	4-item structured rubric (e.g. "were multiple site options discussed?"), scored 0–3 each, rescaled 0–1	Secondary
disagreement_resolved	g — GP	"Were recorded objections addressed before the plan was finalized?" (Y/N)	Secondary

g = measured once per Gram Panchayat (repeats across every household in that GP).
i = measured separately for each household/respondent.

Sample Dataframe

Illustrative — not real data. *One row per household (i), flattened within each GP (g). Teal headers = X_ig controls, gold headers = Y_ig outcomes.*

gp_id	hh_id	T_g	baseAtt% (g)	base_attendance (i)	attendance_share (g)
GP014	HH01	1	0.34	1	0.61
GP014	HH02	1	0.34	0	0.61
GP014	HH03	1	0.34	1	0.61
GP007	HH01	0	0.33	0	0.39
GP007	HH02	0	0.33	1	0.39
GP007	HH03	0	0.33	0	0.39

Notice: every "(g)" column repeats identically across all 3 households within GP014, and again within GP007 — because they are GP-level facts, not household facts. Only the "(i)" columns vary row to row.

The Hypothesis Test

Mapping the dataframe onto the regression:

Regression term	Dataframe column(s)
Y_ig (outcome)	deliberation = z-avg(attendance_share, speaking_diversity_index, deliberation_quality, disagreement_resolved)
T_g (treatment)	T_g
X_ig (controls)	baseAtt%, base_attendance_i + rest of household/GP controls (Methodology, Slide 4)

$$\mathrm{Deliberation}_{ig} = \beta_0 + \beta_1 T_g + \boldsymbol{\beta}_2' X_{ig} + \varepsilon_{ig}$$

Feeding the dataframe in gives (illustrative, not real data):

Outcome	β_1 (est.)	SE	p
Deliberation_index (primary)	**0.31**	**0.10**	**0.002**

Outcome Variables and Index Construction

Variable (Y)	Level	Question asked / how data is picked up	Role
satisfaction_outcome	**i — Household**	**"Overall, how satisfied are you with what was ultimately approved for your village this year?"**	**Primary**
satisfaction_process	i — Household	"How satisfied are you with how this year's planning process was conducted?"	Secondary
perceived_fairness	i — Household	"How fair was the distribution of approved works across different groups in the village?"	Secondary
would_repeat	i — Household	"Would you want this same planning process used again next year?" (Y/N)	Secondary

g = measured once per Gram Panchayat (repeats across every household in that GP).
i = measured separately for each household/respondent.

Sample Dataframe

Illustrative — not real data. *One row per household (i), flattened within each GP (g). Teal headers = X_ig controls, gold headers = Y_ig outcomes.*

gp_id	hh_id	T_g	mgnrega_util (g)	land_ha (i)	Satisfaction_outcome (i)
GP014	HH01	1	0.71	2.1	4.1
GP014	HH02	1	0.71	0.9	3.8
GP014	HH03	1	0.71	3.4	4.3
GP007	HH01	0	0.58	1.5	3.2
GP007	HH02	0	0.58	2.0	3.0
GP007	HH03	0	0.58	0.8	3.4

Notice: every "(g)" column repeats identically across all 3 households within GP014, and again within GP007 — because they are GP-level facts, not household facts. Only the "(i)" columns vary row to row.

The Hypothesis Test

Mapping the dataframe onto the regression:

Regression term	Dataframe column(s)
Y_ig (outcome)	satisfaction = z-avg(satisfaction_outcome, satisfaction_process, perceived_fairness, would_repeat)
T_g (treatment)	T_g
X_ig (controls)	mgnrega_util, land_ha + rest of household/GP controls (Methodology, Slide 4)

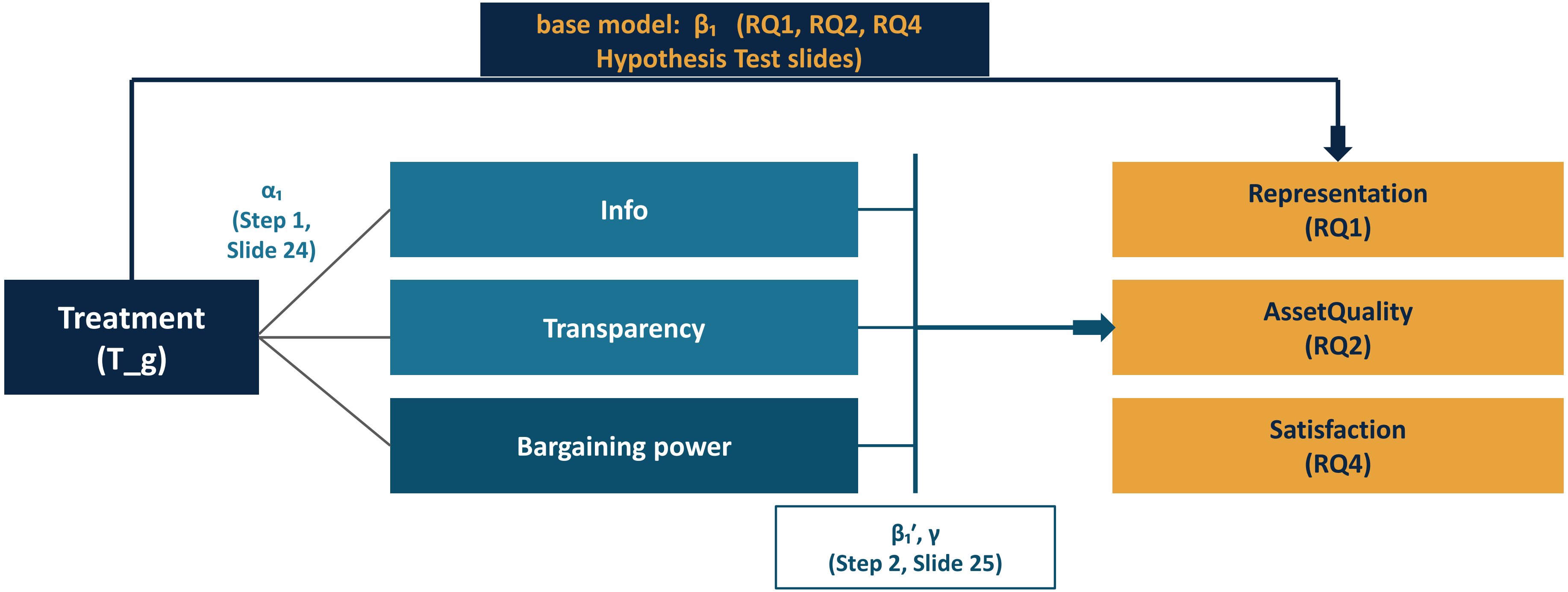
$$\text{Satisfaction}_{ig} = \beta_0 + \beta_1 T_g + \boldsymbol{\beta}_2' X_{ig} + \varepsilon_{ig}$$

Feeding the dataframe in gives (illustrative, not real data):

Outcome	β_1 (est.)	SE	p
Satisfaction_index (primary)	**0.24**	**0.09**	**0.008**

What Is RQ5 Actually Evaluated On?

- RQ5 splits each RQ's base-model arrow (β_1) into a two-stage mediator path — treatment first moves a mediator (α_1), which then predicts the outcome (β_1', γ).



$$\beta_1 = \underbrace{\beta_1'}_{\text{direct}} + \underbrace{\alpha_1^{\text{Info}}\gamma_1}_{\text{via Info}} + \underbrace{\alpha_1^{\text{Transp}}\gamma_2}_{\text{via Transp.}} + \underbrace{\alpha_1^{\text{Barg}}\gamma_3}_{\text{via Bargaining}}$$

The base-model arrow (β_1, top) and the two-stage path (α_1 then β_1'/γ, through the mediators) are not two competing estimates — this identity is what ties them together exactly.

Measuring the Three Channels

Information

- **NEW variables — a knowledge test.**
- Objective: baseline + endline quiz (e.g. "has groundwater gone up/down/same", "how many structures exist"), scored for accuracy.
- Subjective: "how well do you understand local water conditions?" (5-pt)
- info_index = z-score average of both.

Transparency

- **NEW variables — is the process legible, distinct from learning facts.**
- "Were reasons for selecting/rejecting a work explained?" (Y/N/partial)
- "Did you see data/maps used to justify the decision?" (Y/N)
- "If rejected, were you told why?" (Y/N/NA)

Bargaining Power

- **REUSED from RQ3 — this is literally what deliberation quality measures.**
- attendance_share
- speaking_diversity_index
- deliberation_quality (observer rubric)

Step 1: Does Treatment Move Each Mediator?

- Necessary condition for mediation: a channel can't explain an effect unless treatment actually moves it. Each mediator is regressed on treatment separately, exactly like an RQ1–4 outcome:

$$M_{ig}^{\text{Info}} = \alpha_0 + \alpha_1^{\text{Info}} T_g + \boldsymbol{\alpha}_2' X_{ig} + u_{ig}$$

$$M_{ig}^{\text{Transp}} = \alpha_0 + \alpha_1^{\text{Transp}} T_g + \boldsymbol{\alpha}_2' X_{ig} + u_{ig}$$

$$M_{ig}^{\text{Barg}} = \alpha_0 + \alpha_1^{\text{Barg}} T_g + \boldsymbol{\alpha}_2' X_{ig} + u_{ig}$$

α_1 in each equation is analogous to β_1 in RQ1–4 — it's the treatment effect on that mediator specifically. If, say, α_1^Info ≈ 0, the information channel can be ruled out immediately, without needing Step 2.

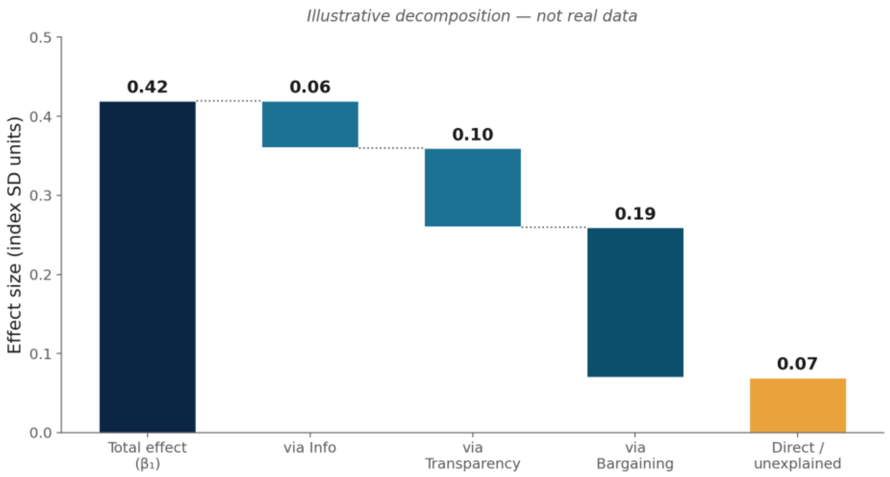
Step 2: The Multiple-Mediator "Horse Race"

- All three mediators are entered together into the outcome regression (run separately for each of RQ1/RQ2/RQ4's index outcomes):

$$Y_{ig} = \beta_0' + \beta_1' T_g + \gamma_1 M_{ig}^{\text{Info}} + \gamma_2 M_{ig}^{\text{Transp}} + \gamma_3 M_{ig}^{\text{Barg}} + \boldsymbol{\beta}_2' X_{ig} + \varepsilon_{ig}$$

- β_1' (direct effect, controlling for all three mediators) vs. β_1 (total effect from RQ1/2/4). The gap is decomposed into each channel's contribution via the product of coefficients:

$$\beta_1 = \underbrace{\beta_1'}_{\text{direct}} + \underbrace{\alpha_1^{\text{Info}}\gamma_1}_{\text{via Info}} + \underbrace{\alpha_1^{\text{Transp}}\gamma_2}_{\text{via Transp.}} + \underbrace{\alpha_1^{\text{Barg}}\gamma_3}_{\text{via Bargaining}}$$



Step 3: Do Channels Reinforce Each Other?

- We treat channels as additive. But a natural question — does transparency only matter when bargaining power is also higher — needs an interaction term:

$$Y_{ig} = \beta_0'' + \beta_1'' T_g + \gamma_1 M_{ig}^{\text{Transp}} + \gamma_2 M_{ig}^{\text{Barg}} + \gamma_3 \left(M_{ig}^{\text{Transp}} \times M_{ig}^{\text{Barg}}\right) + \boldsymbol{\beta}_2' X_{ig} + \varepsilon_{ig}$$

- γ_3 significant and positive: the two channels compound — transparency matters more when a group also has more room to act on it, and vice versa.
- $\gamma_3 \approx 0$: channels operate independently; their separate coefficients (γ_1, γ_2) already tell the full story.
- *This linear product-of-coefficients approach is a simplified, teachable version of mediation analysis. For a rigorous treatment with formal identification assumptions and confidence intervals on indirect effects, see Imai, Keele & Tingley (2010), "A General Approach to Causal Mediation Analysis."*

The Full Pipeline, in One Picture

1. Research question → 2. Outcome variables → 3. Standardize → index → 4. Regress on treatment (β_1) → 5. Correct for multiple testing → 6. (RQ5 only) mediation decomposition

- This pipeline is identical for RQ1–RQ4; RQ5 adds one more stage on top, reusing the same variables and the same regression machinery rather than inventing a parallel analysis.

Discussion

- RQ4's illustrative results show satisfaction moving even where an underlying outcome might not (see the RQ2/RQ4 comparison). If that pattern held in real data, how would you interpret it — and would it change your recommendation to FES?
- The info_index depends on costly independent field verification. If budget forced a cap on the number of GPs in the sample (statistical power), how would you still develop an understanding of the underlying impact pathway?
- Suppose the direct/unexplained share in RQ5 came out large (e.g. 50%+) for every outcome. What would that suggest about the three channels this study chose to measure?